

5 October 2026

Dear Editor-in-Chief,

We submit "Innovation-Normalized Detection in Compound-Gaussian Clutter: Exact Laws, Conformal Thresholds and Certified Integration under Pulsed Interference" as a Regular Paper in IEEE Transactions on Radar Systems. Authors, in order: Muhammad Umair Raza, Sohail Ahmed, Ammad Ahmed, M. Atif Shahzad and Syed M. Kazam Abbas Kazmi.

Under a true AR compound-Gaussian model with a common episode texture and no thermal noise, whitening and innovation normalization recover the classical CA and OS laws. We derive a post-onset detection law, its self-masking horizon and the delay provided by a slow-time guard. For exchangeable clean episodes, a conformal threshold on a worst-case bound certifies clipped and binary integration against arbitrary-amplitude interference under a dominating, clutter-independent hit model. These are conditional contributions; general empirical superiority is not claimed.

Measured validation uses IPIX and NetRAD sea clutter, JKU 77 GHz FMCW recordings and camera-labeled University of Washington radar recordings. On IPIX, conformal thresholds keep twelve of thirteen statistics within 0.73–1.47 times design down to  $10^{-4}$ . On NetRAD, excluded from initial development, refitted thresholds stay within 1.36 times design at  $10^{-3}$ ; five statistics exceed twice design at  $10^{-4}$ . The per-look score shares the PAMF's whitening gain.

Three prospectively registered studies strengthen the evidence and delimit its scope. Injected known-onset trajectories through 48 looks confirm delayed, eventual collapse. The guarded screen acquires only 0.004–0.030 across all 24 idealized continuous-migration conditions; P-ANMF has a higher point estimate in every condition. All 3 camera-labeled recordings fail the correlation gate and provide no primary clean-entry units. Their uncertain depth and timing support descriptive presence association, with no demonstrated whitening advantage. Independent, precisely timed real-target onsets in correlated clutter remain unvalidated.

The supplement retains proofs, uncertainty, failures and protocol deviations, including burst-interference and sensitivity limits. Code, protocols and frozen numerical logs are available at <https://github.com/razaumair2203-ux/inarc> (tag r22-final-submission). AI assistance is disclosed in the Acknowledgment.

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Sincerely,

Muhammad Umair Raza, corresponding author, on behalf of the authors

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